Highlights

- Fault classifiers above 0.97 macro-F1 lose 0.11–0.32 at unseen rotation speeds
- A leakage-free, speed-held-out protocol for MAFAULDA and COMFAULDA
- A random forest on order features is the most accurate and stable model overall
- A speed-conditioned Fourier neural operator is best for unbalance (0.93 vs 0.85)
- Compound faults are not recognised from single-fault training (exact match 5%)